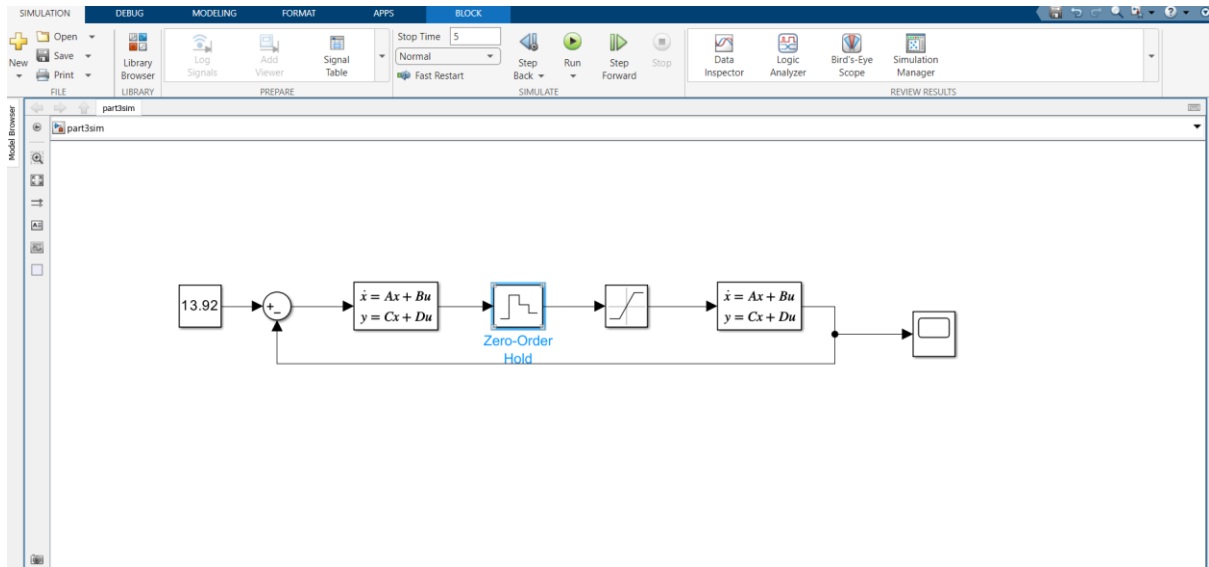


PE PROJECT PART 3

From part 2, we have steady state: 13.92

Settling time initially chosen (at -2% of steady state): 8.8 ms (0.0088 s)



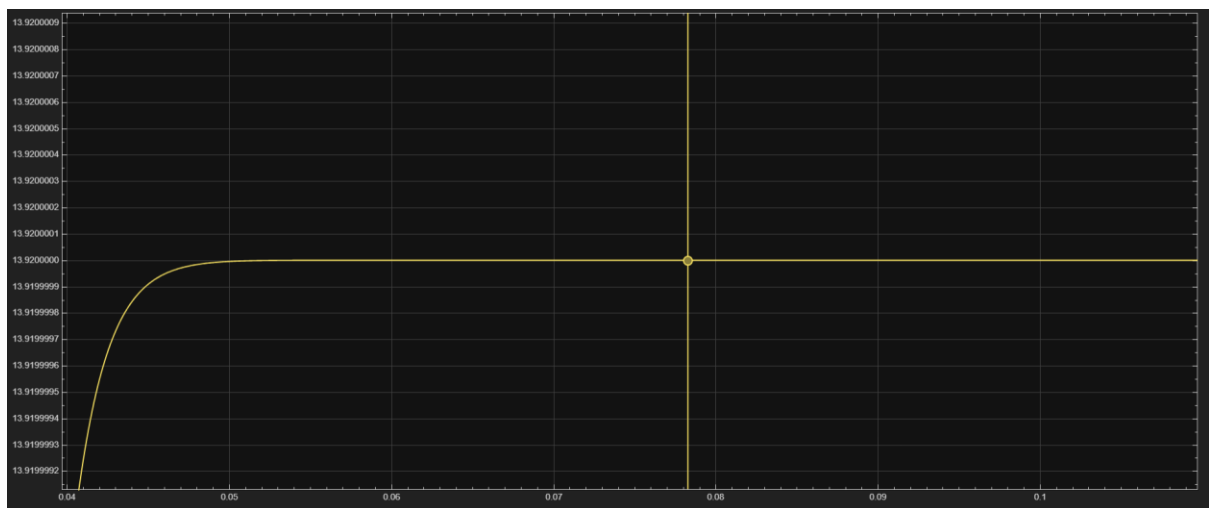
The second transfer function H_o (with state space) is our defined tf with 0 overshoot, chosen settling time from the previous part, damping factor of 1 and further calculated ω_n .

The first transfer function is the controller H_c , found using the formula $(1/H_p) * (H_o / (1 - H_o))$. I used `ssdata` to get the state space values from H_c .

„Zero-Order Hold” period set to 1/f/10 (or PWM period/10) – sampling it 10x faster



Statistics							
State-Space	Max	Min	Peak to Peak	Mean	Standard Deviation	Median	RMS
Time	56.4000 ms	0.0000 s					
Value	13.9200000013225	0	13.9200	13.7667	1.0904	13.9200	13.8298

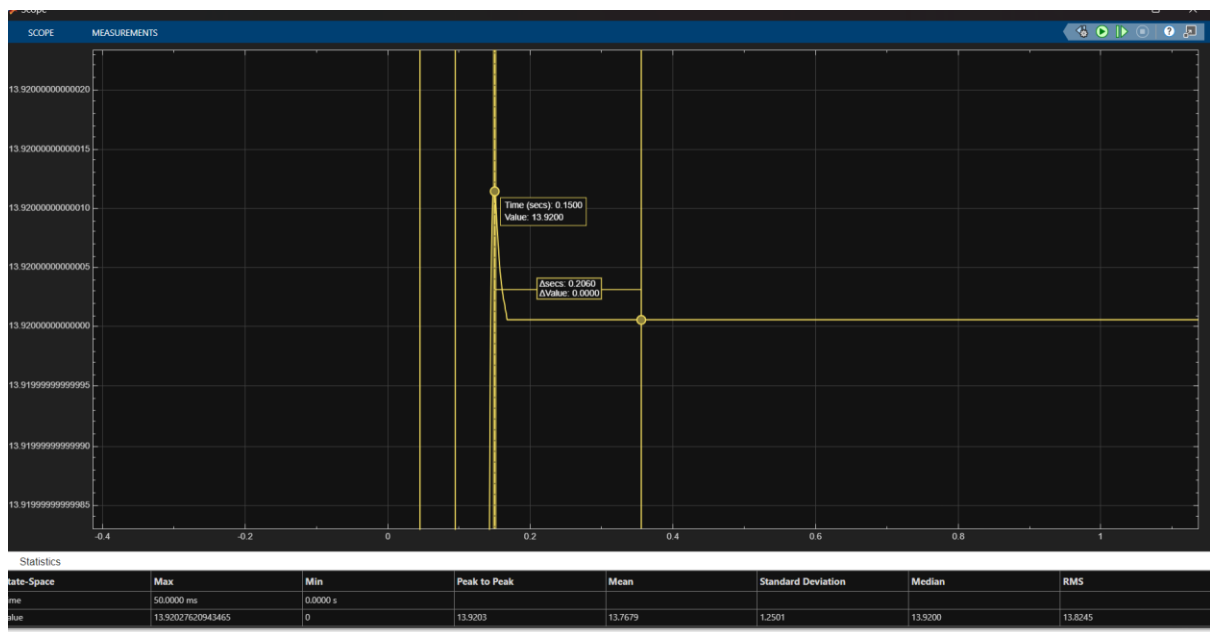


Statistics							
State-Space	Max	Min	Peak to Peak	Mean	Standard Deviation	Median	RMS
Time	56.4000	39.7333					
Value	13.9200000013225	13.9199854916956	0.0000	13.9200	0.0000	13.9200	13.9200

Max value: 13.92000..

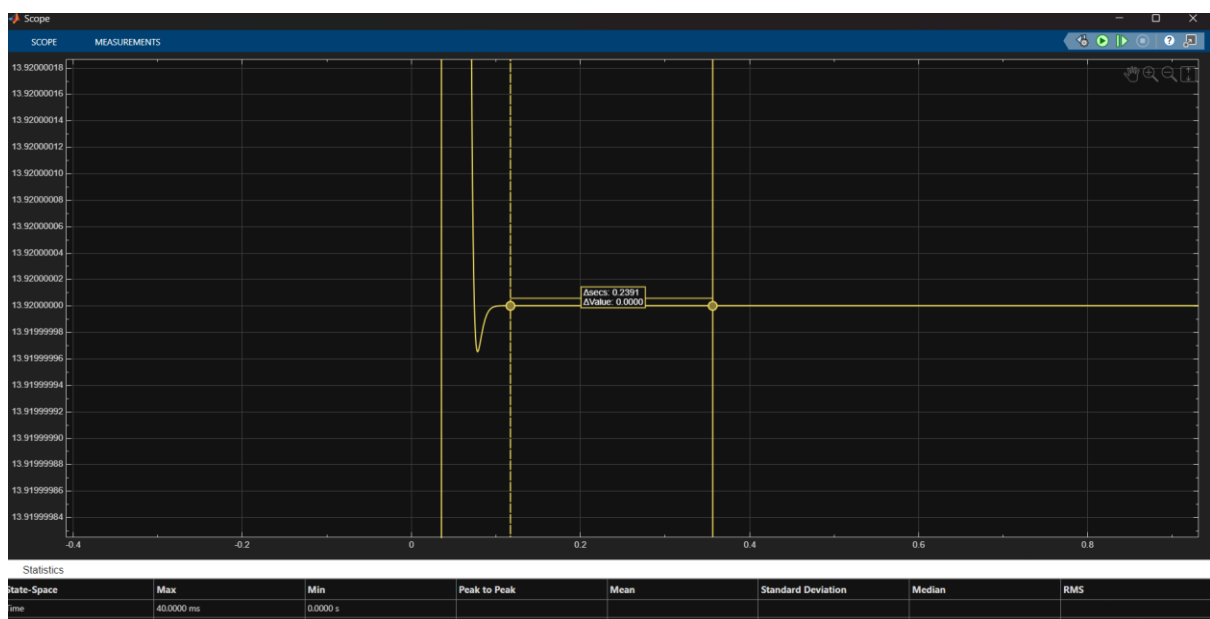
(slight very very small overshoot)

„Zero-Order Hold” period set to $1/f$ (or PWM period)



There is an obvious overshoot now and since the controller only updates once every PWM cycle, the solver struggles and has many discontinuities and visual issues. (although I encountered a problem which was that a few hours ago my scope didn't show these dramatic 'spikes', now they don't go away)

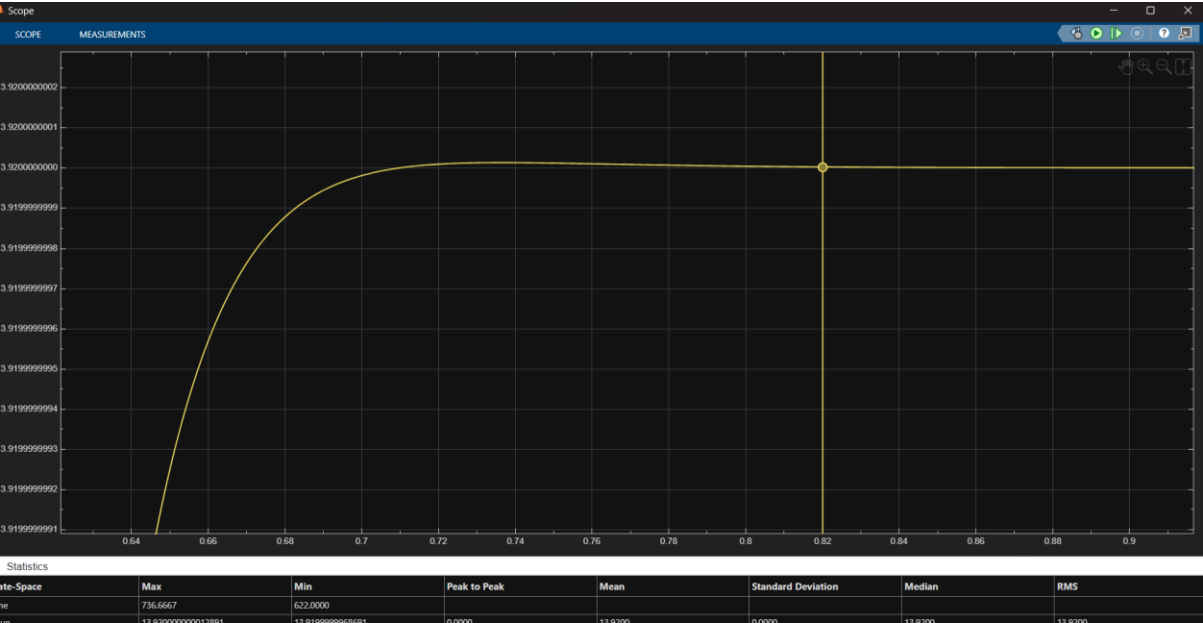
Changed $t_s = 15$ ms



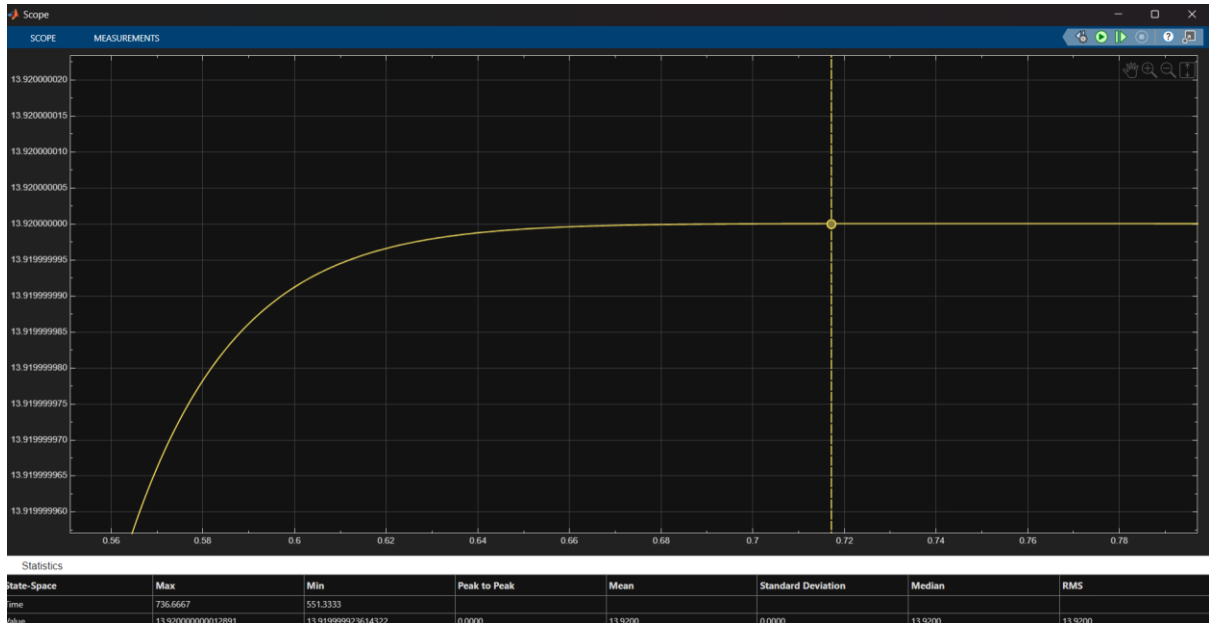
$T_s = 0.0880\text{ s}$



$T_s = 0.1056\text{ s}$ (getting better)



$$T_s = 0.1056 \text{ s } (12 \cdot 8.8 \text{e-}3)$$



Here there seems to be no overshoot.

As a conclusion, if the controller updates slowly (sampling 10 times slower), the output has overshoot because of a slow reaction so we chose a bigger settling time to restore the critically damped behaviour with no overshoot.